

3ai 1	Interference pattern of Black and White equally spaced stripes 	A1
3ai 2	Fringe separation of the interference pattern will double. The bright fringes will be less bright.	A1 A1
3bi	Distance from lower slit to P = $\sqrt{1 + 0.23^2}$ Distance from higher slit to P = $\sqrt{1 + 0.03^2}$ Path difference at point P $= \sqrt{1 + 0.23^2} - \sqrt{1 + 0.03^2}$ $= 0.0257 \text{ m}$	M1 M1 A0
3bii 1	Phase difference = $2.57 / 1.03 \times 2\pi$ $= 5\pi$	M1 A1
3bii 2	Destructive interference , minima observed	A1

3biii	0π	Bright	M1
	1π	Dark	
	2π	Bright	
	3π	Dark	
	4π	Bright	
	5π	Dark	
	3 low intensity regions (count from the table above).		A1
Alternative Method			
Fringe separation = $\lambda D / a = 5.15 \text{ cm}$ M1			
Number of low intensity regions = $13 / 5.15 = 2.52 = 3$ (rounded up) A1			
3c	<u>Central white fringe</u> as all frequencies of light will overlap.		A1
	<u>Continuous bright coloured spectrum and dark fringes on either side,</u> <u>Due to different wavelengths of light having different fringe separations.</u>		A1 M1
4ai	Maximum KE		